Robust Optimal Transport

1 Introduction

This note studies robust optimal transport when support points are uncertain. The adversary moves each point inside a ball. unsupervised pretraining is not the focus of this note.

2 Method

We pre-commit to a transport plan before the perturbation.

The resulting problem is a convex-concave minimax problem.

Each support point shares one displacement across its edges.

3 Experiments

We compare against nominal optimal transport on synthetic data.

The robust model reduces the worst-case cost substantially.